

Arabic Quality Platform

UML Design Document

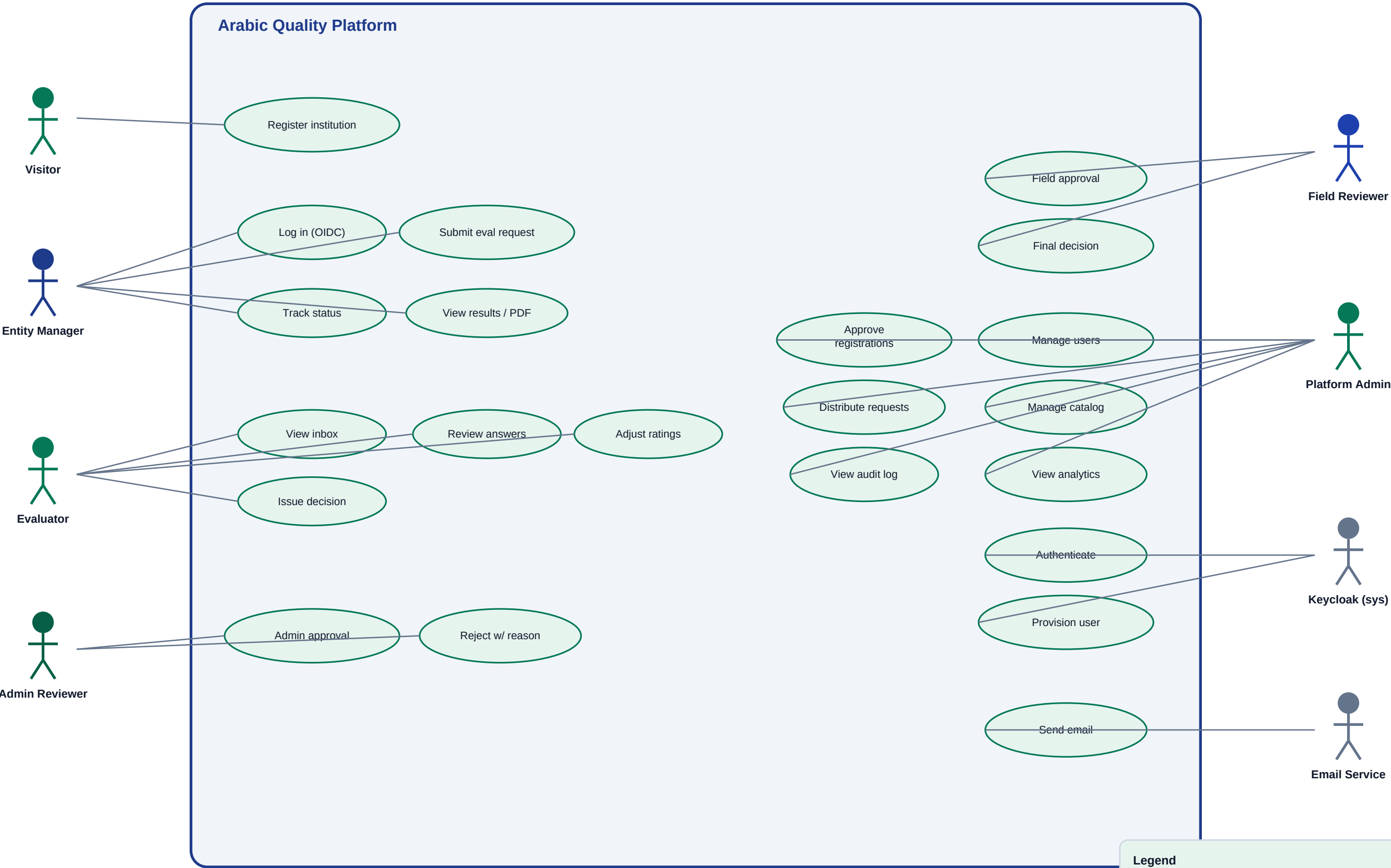
Quality assessment of Arabic teaching for non-native speakers

Diagrams included

1. Use Case Diagram
2. Class Diagram (Domain Model)
3. Sequence Diagram - Registration Approval Workflow
4. State Machine - Evaluation Request Lifecycle
5. Architecture Diagram (3-tier + Keycloak)
6. Entity-Relationship Diagram (ERD)
7. Deployment Diagram (Docker Containers)
8. Activity Diagram - Full Evaluation Cycle

1. Use Case Diagram

Actors and their interactions with the platform



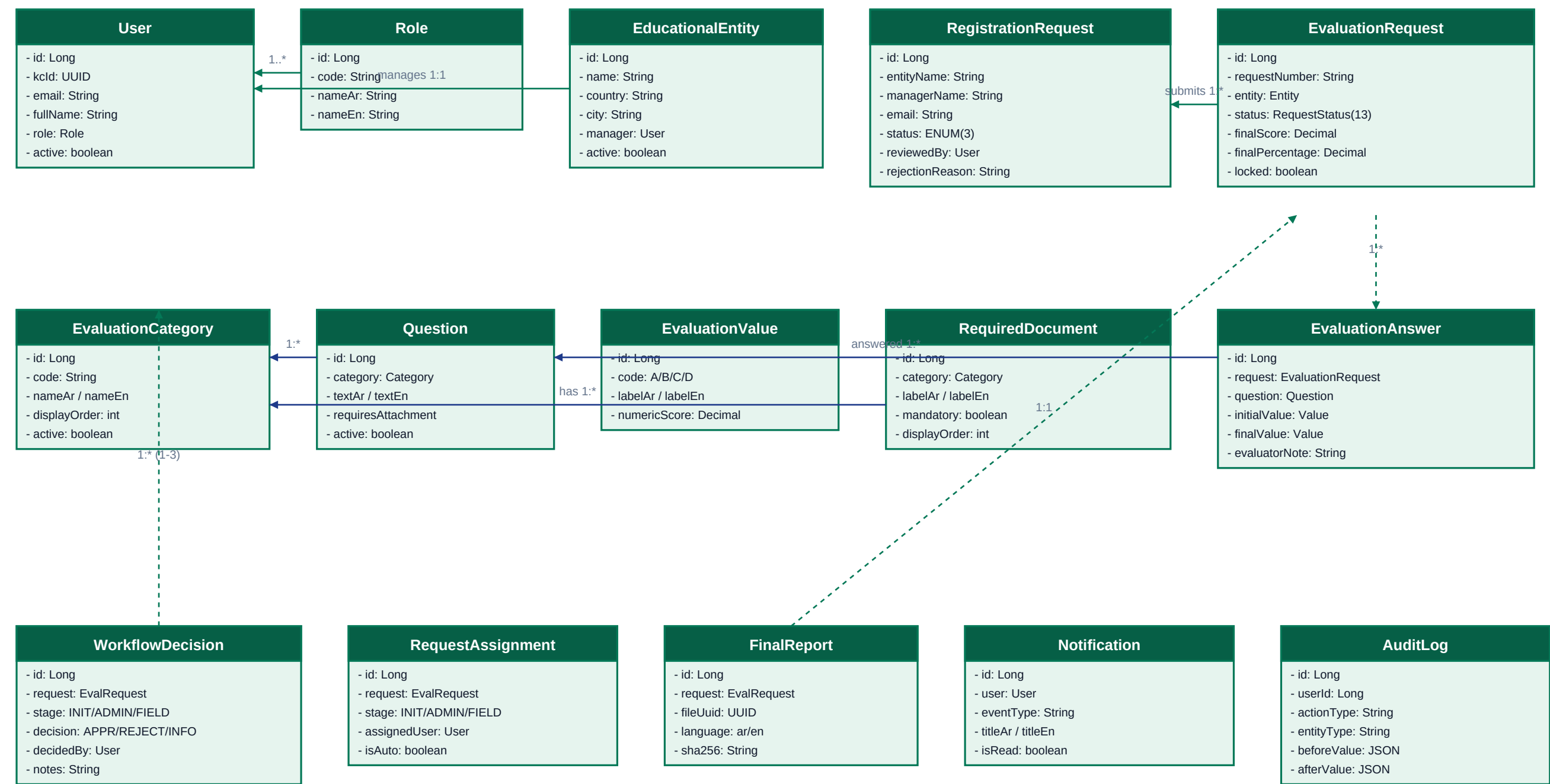
Legend

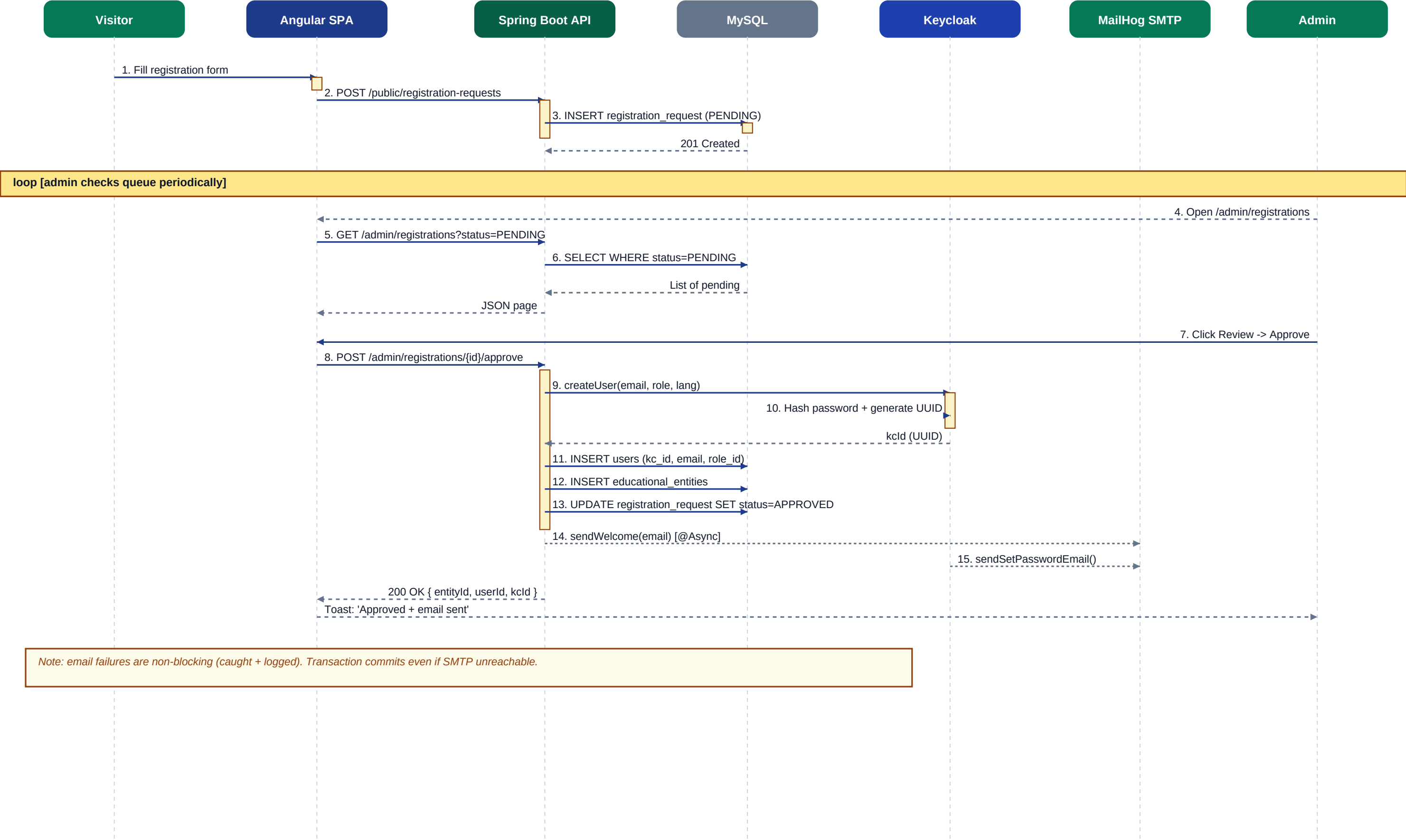
Stick figure = Actor (human or system)

Ellipse = Use case

2. Class Diagram

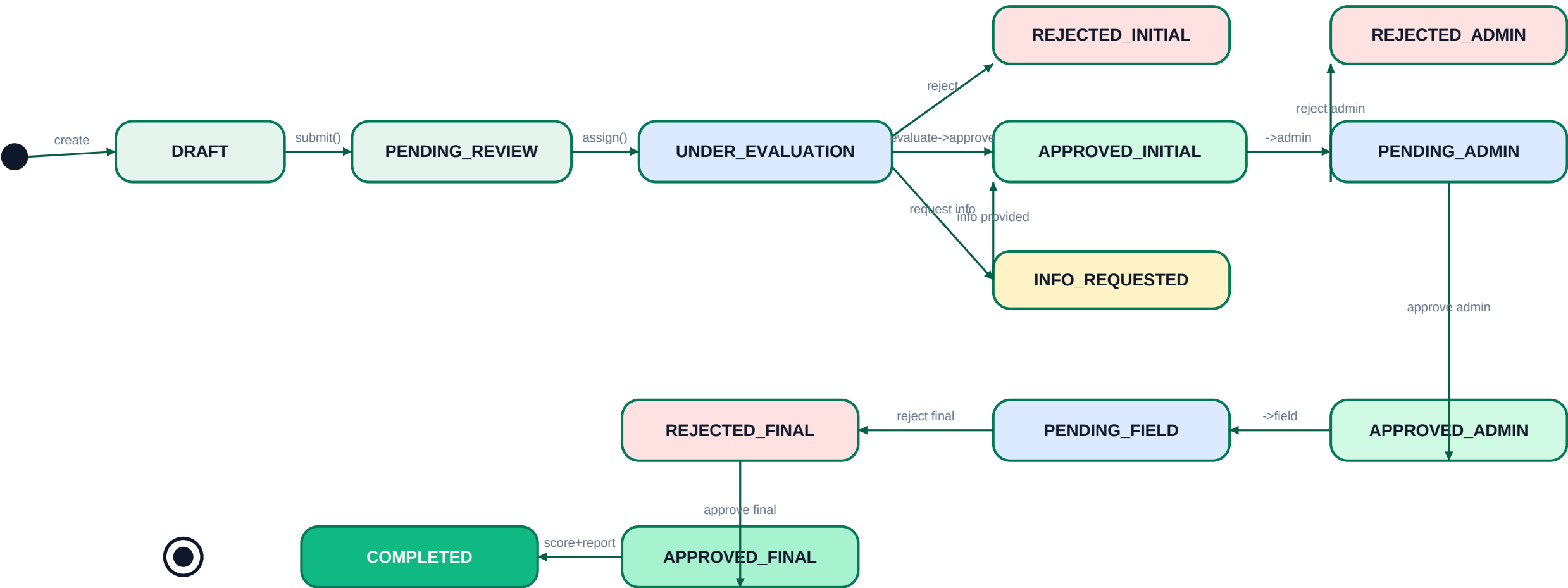
Core domain entities and their relationships





4. State Machine Diagram

Lifecycle of an EvaluationRequest - 13 states governed by role-based transitions



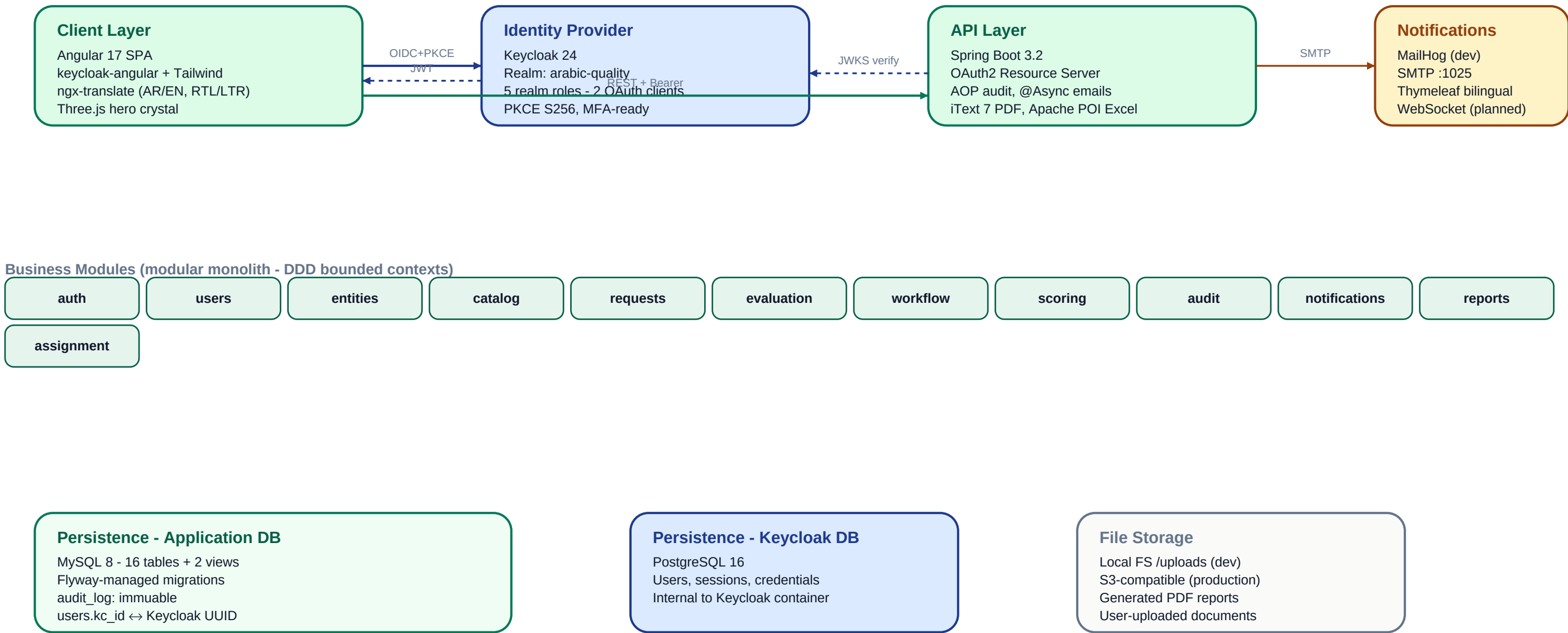
Legend

DRAFT/PENDING/UNDER_EVALUATION - In-progress states
APPROVED_INITIAL/ADMIN - Successful intermediate transitions

- COMPLETED - Report available, request locked
- REJECTED_* - Terminal failure (can re-submit)

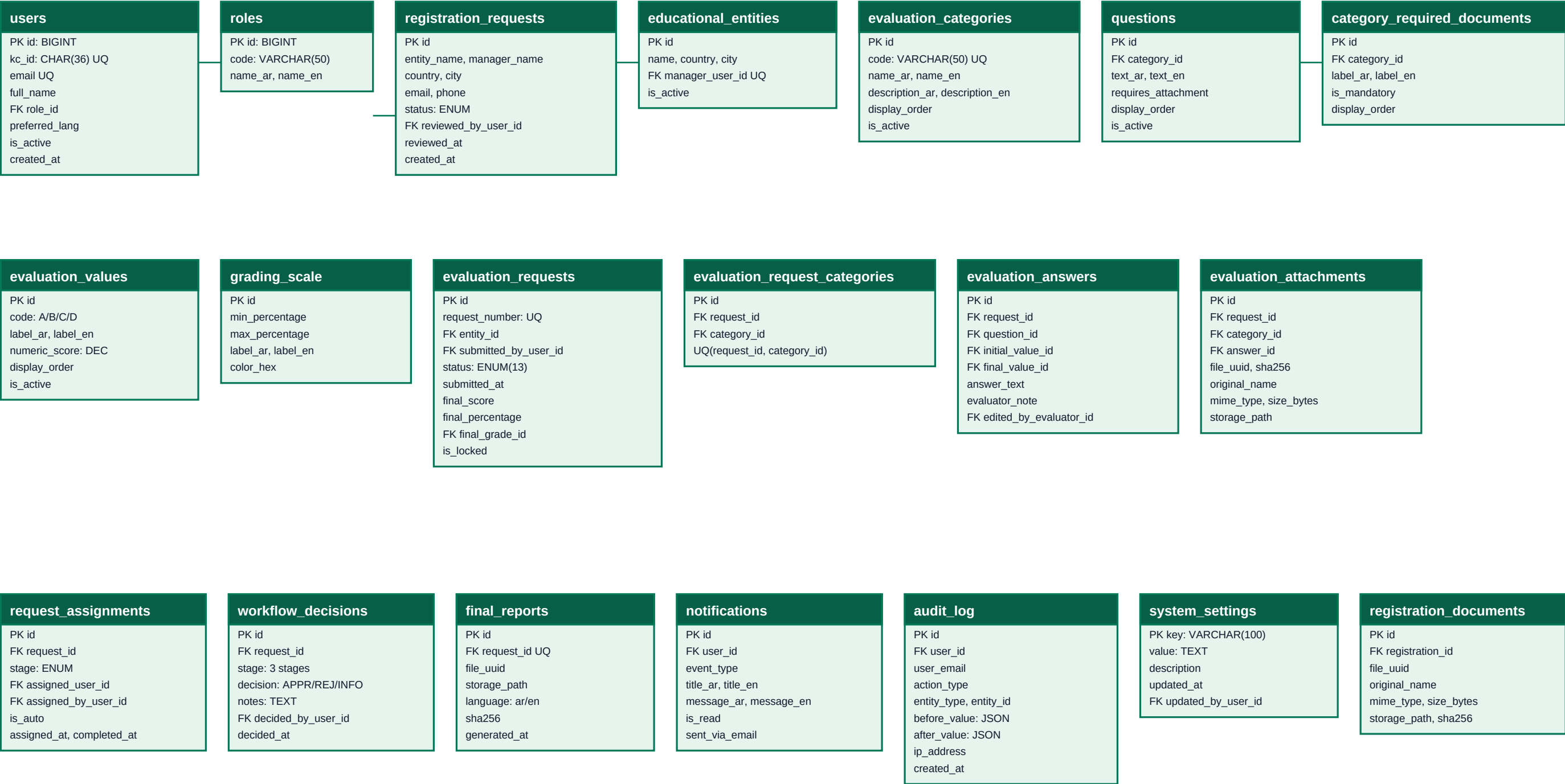
5. Architecture Diagram

3-tier modular monolith with Keycloak delegation (OAuth2/OIDC)



6. Entity-Relationship Diagram (ERD)

16 tables + 2 views - Flyway-managed - Audit-grade traceability

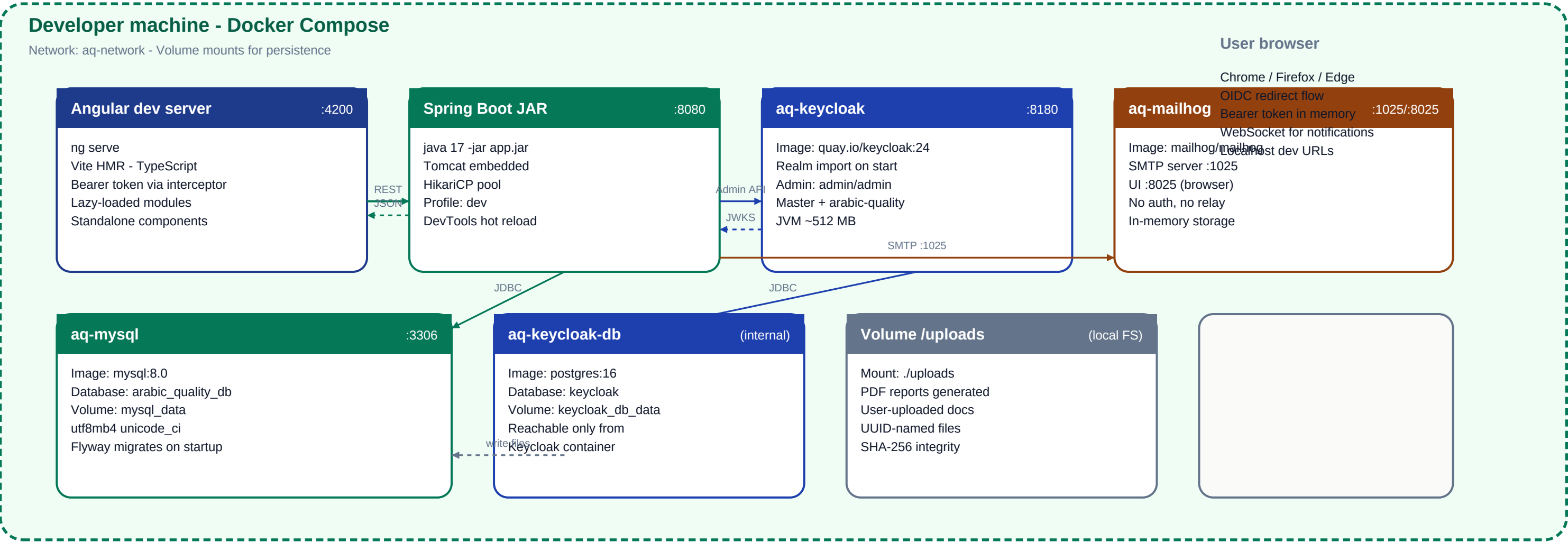


Legend

PK = Primary key FK = Foreign key
UQ = Unique constraint ENUM = MySQL enum

7. Deployment Diagram

Docker Compose dev stack + production target topology



Production target topology

Nginx (TLS + reverse proxy) -> Angular static (CDN) - Spring Boot JAR (PM2/systemd) - Keycloak cluster
MySQL managed (RDS/CloudSQL) - PostgreSQL for Keycloak - S3-compatible storage - SMTP via SendGrid/Mailgun - CI/CD via GitHub Actions

8. Activity Diagram

Full evaluation cycle - from public registration to official report issuance

